Lead Solo Workshop

An improvisation guide for adding lead solos to a song: a six-step plan, eight techniques, chord-tone targeting, a lick library, a full nine-bar solo and backing tracks. Every lick opens on the fretboard map, which gains boxes, chord-tone rings and a lick layer.

Built on what the app already has

`ScaleFretmapFloat` and `ScaleFretmapNeck` for the popup, `playLead()` and `articulation()` for bends, slides and vibrato, `noteToken()` and `layoutTab()` for TAB, and the Theory player for backing tracks and highlights. Licks are plain `PlayableNote`s, so nothing new has to understand a different note format.

Files that come with it

- `cursor-prompt.md` : the build instructions
- `soloGuide.ts` : all content and helpers (boxes, chord tones, lick overlay, backing, examples)
- `soloGuide.test.ts` : 14 tests; with them the full frontend suite passes (139) and type-checks
- `reference-prototype.html` : open in a browser to see and hear the behaviour

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Prototype screenshots, 25 October 2026

1 · WHAT CHANGES

Extend the fretboard map; add one page

Everything else is reuse. No backend changes.

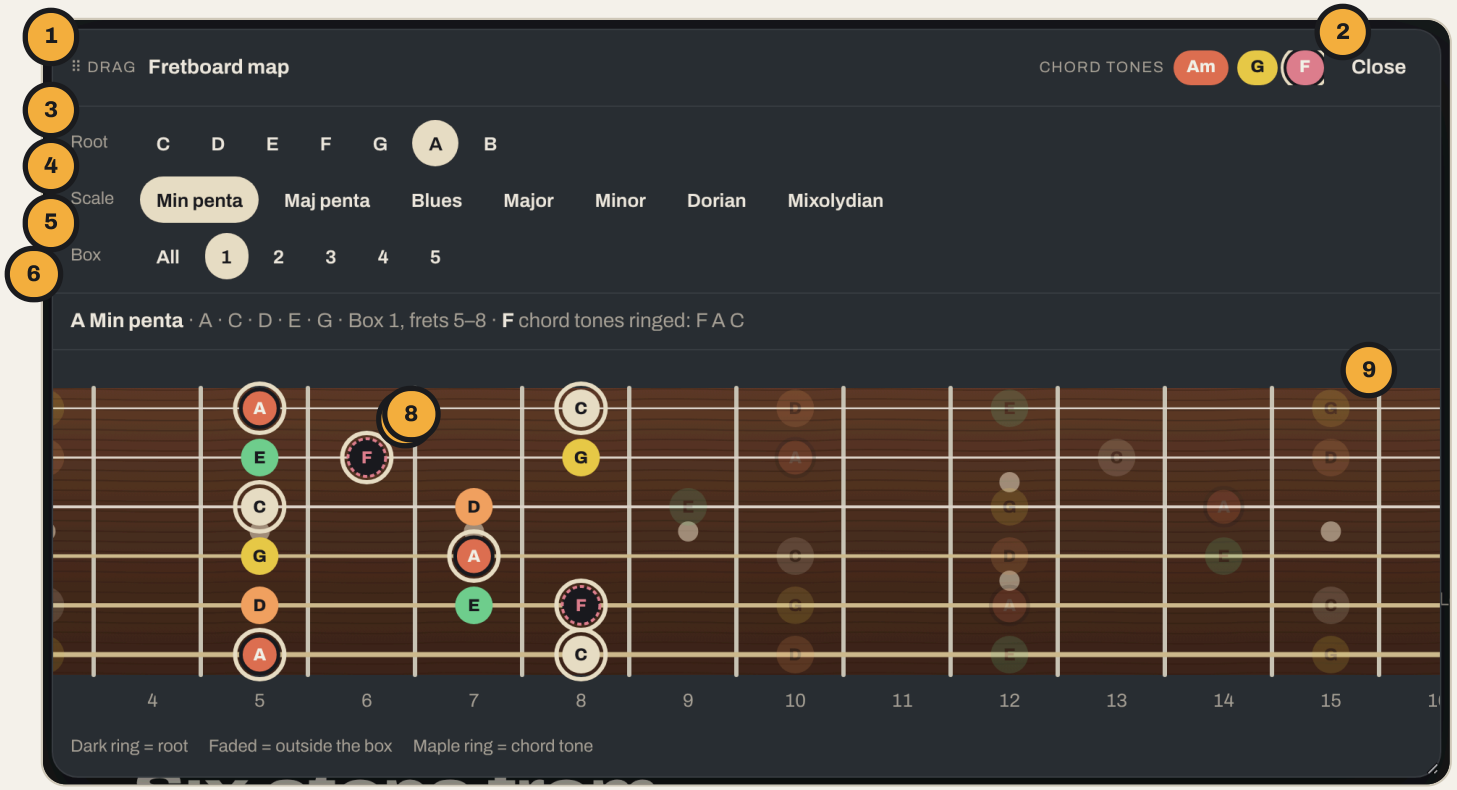
PIECE	HAS TODAY	ADDS
`ScaleFretmapNeck`	Scale notes; chord-shape finger placement	Optional props: `box` (fade notes outside it), `chordTones` (maple rings, dashed markers for non-scale tones), `lick` (markers, order badges, technique glyphs, lit note), `fretWindow` (crop for card diagrams)
`ScaleFretmapFloat`	Drag, Root, Scale, chord pills (shapes), legend, neck	Box row (pentatonic and blues only), chord pills switch to chord tones when opened with a chord list, lick bar (Play/Stop, title, TAB, Clear lick), technique key
Audio	Theory player (backing, highlights); `playLead()` + `articulation()`	Theory player gets an optional lead line: `play(ex, onHighlight, {{ lead, leadTone }})`, plus `loop` for jam tracks
TAB	`noteToken()`, `layoutTab()`, `TabView`	Light the token of the sounding note (`n<i>`)
New	–	`lib/music/soloGuide.ts`, route `/learn/solo`, components in `components/solo/`

Hero title, map button, lead tone	**Plan** 6 numbered steps	**Techniques** 8 cards	**Choose the scale** 6 rows	**Land on chord tones** A/B demo	**Lick library** 8 cards	**Full solo** 9 bars	**Jam** 4 looping tracks

2 · FRETBOARD MAP

Boxes and chord tones

Opened from "Choose the scale", row 1, with F selected. Data: `pentatonicBox()`, `chordToneCells()`.

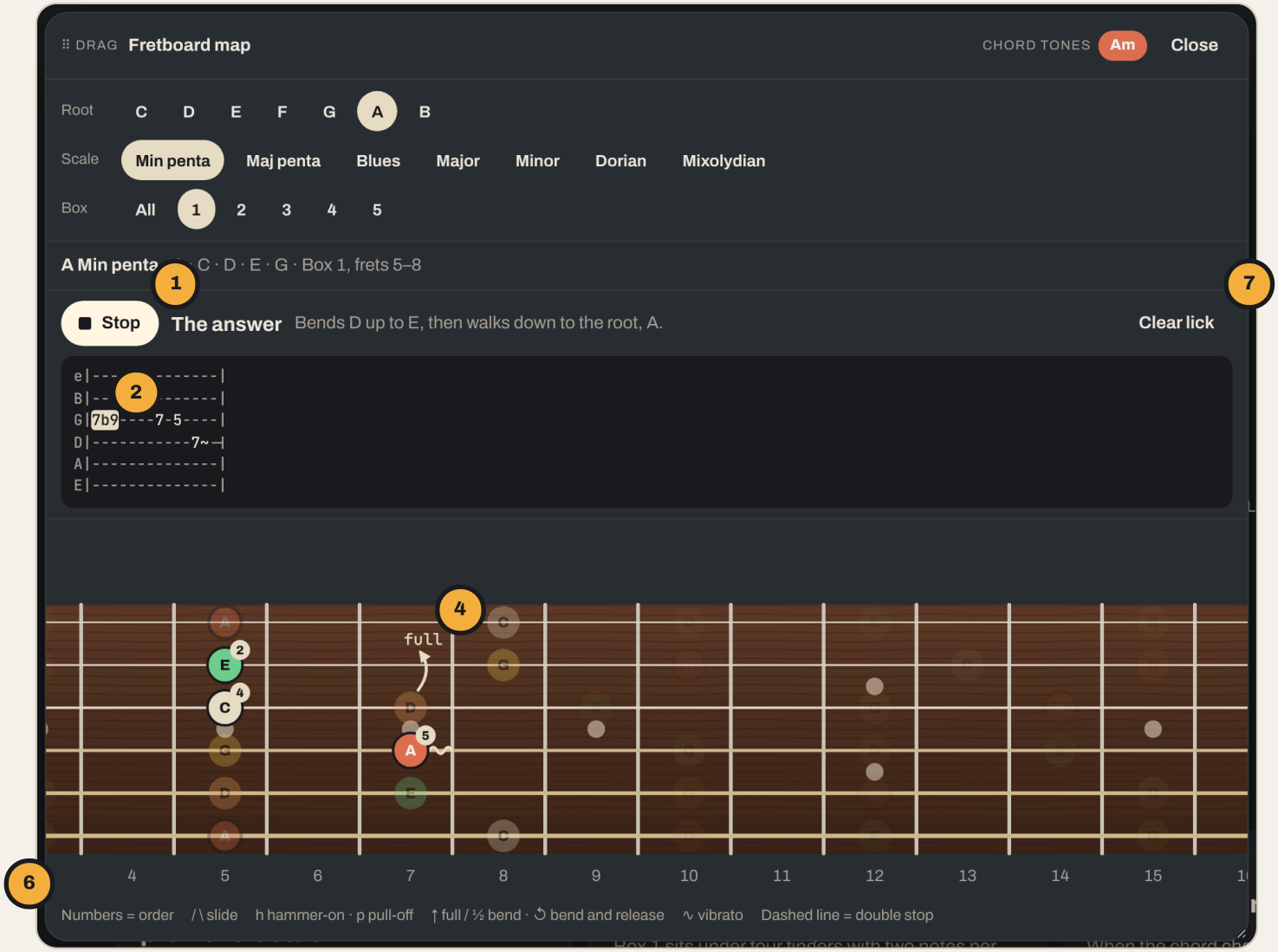


1. **Drag handle**
   Unchanged from `ScaleFretmapFloat`: drag, clamp, remember position.
2. **Chord-tone pills**
   When opened with a chord list, the header pills (coloured by root, `noteMarkerColors`) toggle chord-tone rings instead of chord shapes.
3. **Root**
   Existing `SCALE_ROOTS` row.
4. **Scale**
   Existing `SCALE_KINDS` row. The guide uses min/maj pentatonic, blues, major, minor, Dorian and Mixolydian.
5. **Box (new)**
   All, 1–5. Only for `BOX_MODES`. Box 1 starts on the root on the low E string.
6. **Legend**
   Scale name and notes, then "Box n, frets a–b" and "Chord tones ringed: ...".
7. **Outside the scale**
   Chord tone not in the scale (here F): dark fill, dashed outline in its note colour.
8. **Chord tone**
   Maple ring around the note (same maple as the chord placement ring).
9. **Outside the box**
   Scale notes fade to about 16%.

3 · FRETBOARD MAP

The lick layer

Opened from "The answer" and playing. Data: `lickOverlay(lick.notes)` gives markers and glyphs; highlights come from `lickExample()`.



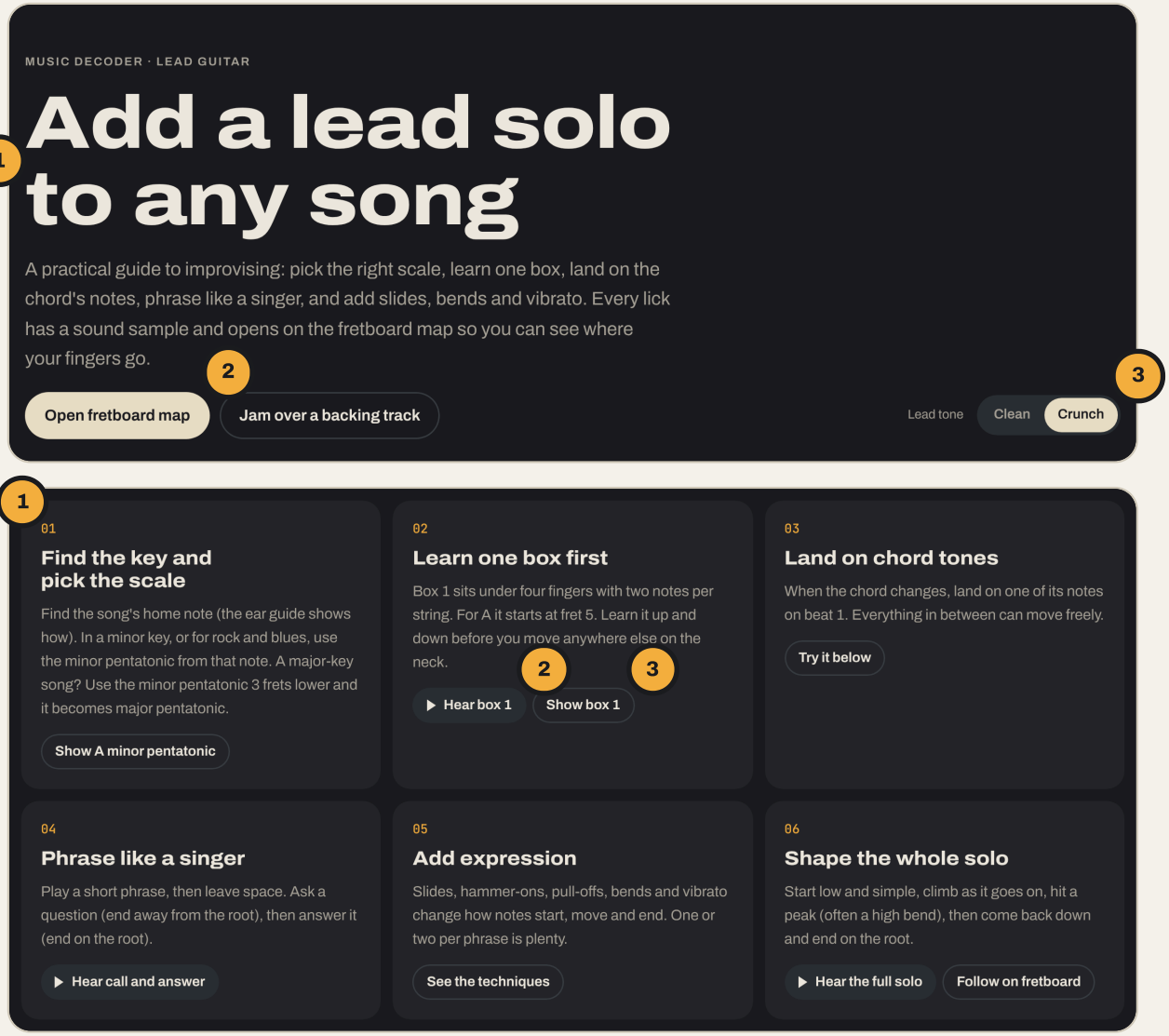
1. **Lick bar**
   Play/Stop (lead + backing together), title and one-line description.
2. **TAB**
   Same tokens as the app's TAB (`noteToken`). The sounding token lights.
3. **Marker**
   Note in its pitch-class colour, maple badge with the play order (first time the position is used).
4. **Technique glyph**
   Bend arrow labelled full or ½, ⟲ for bend and release, slide arrow with / or \, h/p arcs, vibrato wave, dashed link for double stops.
5. **Lit note**
   Maple ring and 1.18× while it sounds. No scaling with reduced motion.
6. **Key**
   One line explaining the glyphs.
7. **Clear lick**
   Back to the plain scale view. Scale notes dim to about 28% while a lick is shown, like the existing chord placement.

For long lines (`lick.compact`, the full solo) draw markers without badges or glyphs and just light the sounding note. When a lick is shown, draw its box in the lick's octave: `pentatonicBox(root, mode, box, lickWindow(notes).from)`.

4 · HERO AND PLAN

Open the map, pick a tone, follow six steps

Data: `PLAN`. Actions are `play`, `playSequence`, `map` or `jump`.



1. **Title**
   Display type (`.display`).
2. **Open fretboard map**
   Opens the popup on A minor pentatonic, box 1, chords Am G F.
3. **Lead tone**
   Clean or Crunch (default). Crunch = the app's lead voice with mild drive; applies to every lick and to the next jam loop.

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1. **Step number**
   Mono, in `--color-next`. A real sequence, so numbered.
2. **Play action**
   Same play/stop button as everywhere.
3. **Map action**
   Opens the popup with that root, scale, box and lick.

5 · TECHNIQUE CARD

Eight cards in two columns

Data: `TECHNIQUES`. Shown: Bend, playing.

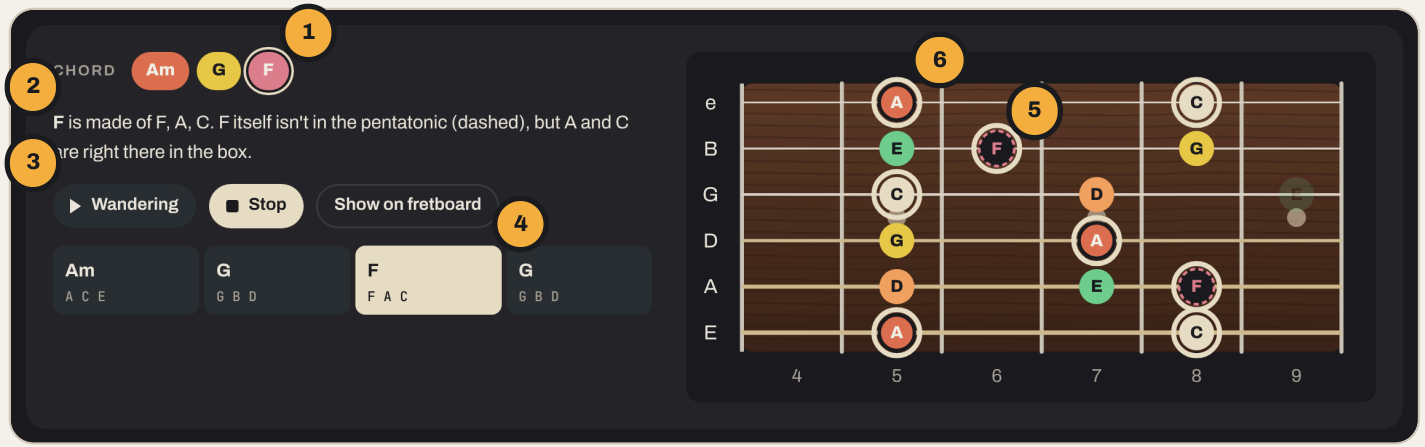


1. **TAB symbol**
   Amber chip (`--color-next`) with the app's notation: 5/7, 5h7, 8p5, 7b9, 8b10r8, 5~, 5+5, rest.
2. **How**
   Three short practical bullets.
3. **Glyph**
   The same lick layer, drawn on a cropped neck (`lickWindow()`, about 6 frets).
4. **Lit note**
   Lights in sync, in the card and in the popup if it shows the same lick.
5. **Hear it**
   Plays the lick (with backing where it has one).
6. **Show on fretboard**
   Opens the popup with this lick, its scale and box, and its backing chords as pills.

6 · CHORD-TONE TARGETING

Pick a chord, see its notes, hear the difference

Data: `TARGET_CHORDS`, `TARGET_TEXT`, `TARGETING`, `WANDERING`.

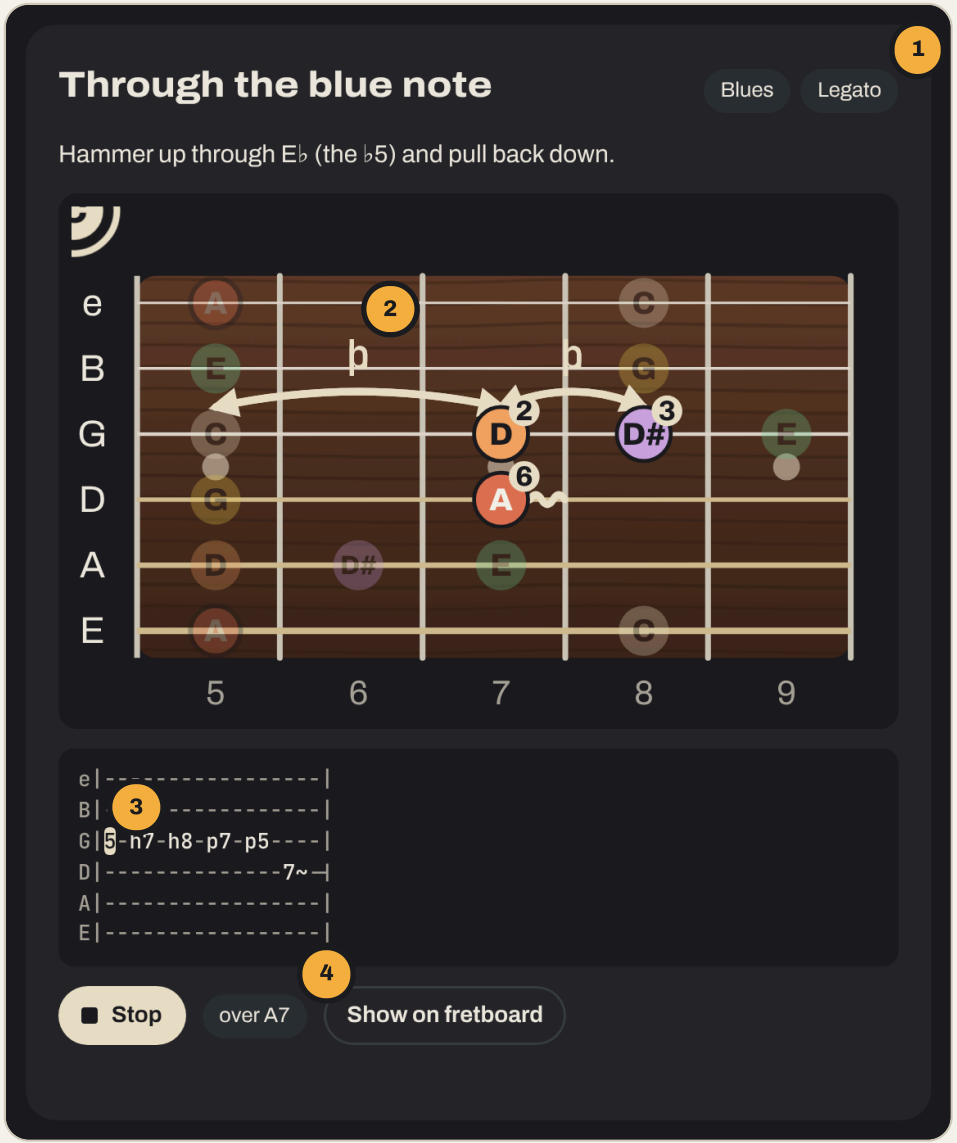


1. **Chord pills**
   Am, G, F. The selected one gets a maple outline.
2. **Explanation**
   Chord notes plus `TARGET_TEXT[chord]`.
3. **A/B**
   Wandering vs Targeting, both over Am – G – F – G.
4. **Bar timeline**
   Chord and its notes per bar; the playing bar lights (`b_<i>`).
5. **Dashed**
   Chord tone outside the scale (F over the F chord).
6. **Ring**
   Chord tone inside box 1.

7 · LICK CARD

Eight licks in two columns

Data: `LICKS`. Shown: "Through the blue note", playing.



1. **Tags**
   Neutral chips (Phrasing, Blues, Legato, ...).

2. **Glyphs**
   h and p arcs on the cropped neck.

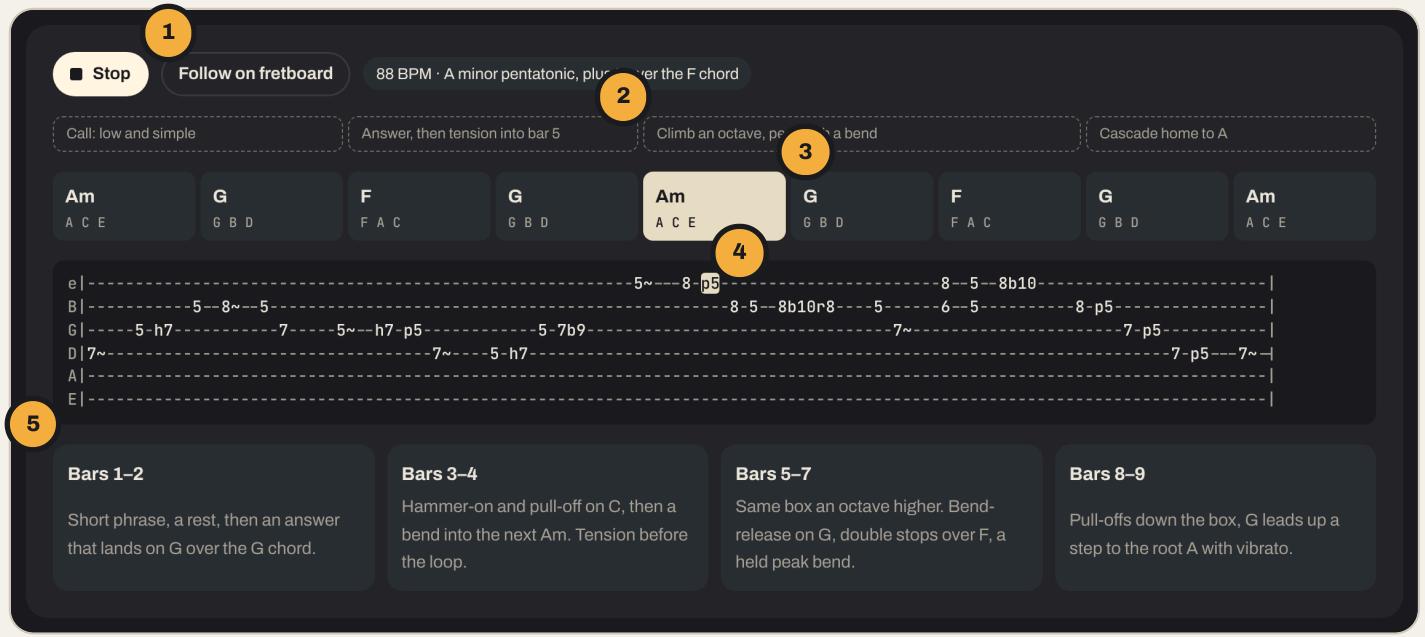
3. **TAB**
   Rendered with the app's TAB layout; the sounding token lights.

4. **Backing chip**
   "over A7": the chord it plays over. Blues licks use the shuffle backing.

8 · FULL SOLO

Nine bars, four sections

Data: `SOLO`, `SOLO_SECTIONS`, `SOLO_NOTES`.

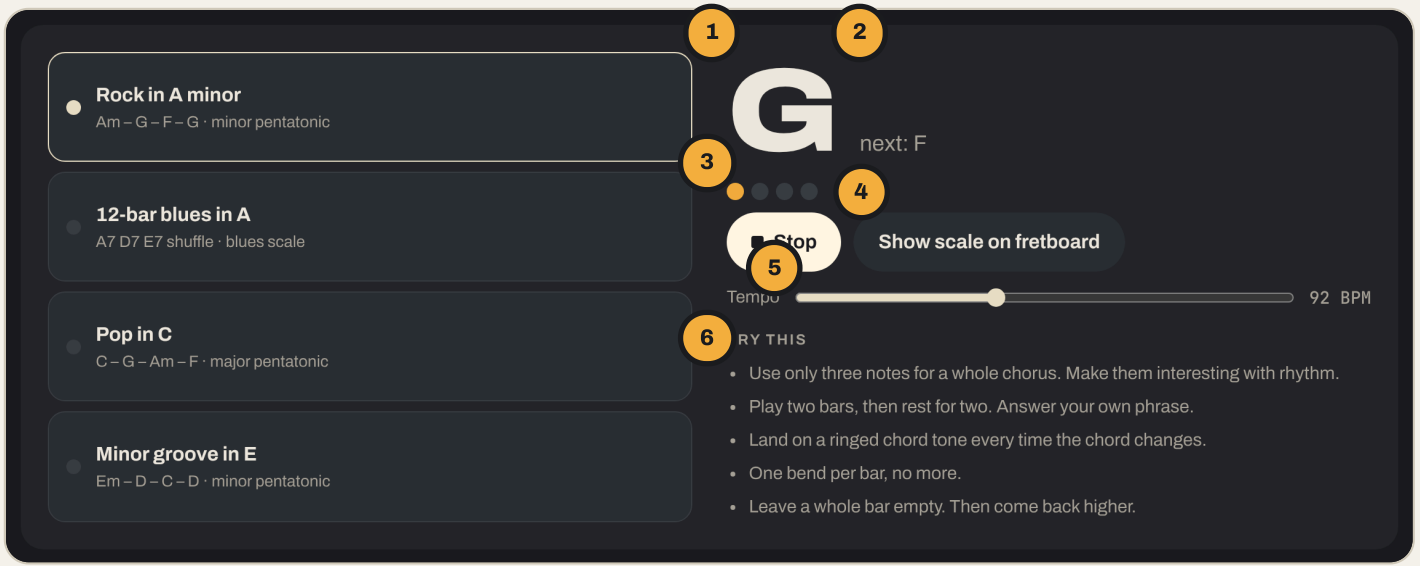


1. **Play**
   Primary button. "Follow on fretboard" opens the popup in compact mode on the whole neck.
2. **Sections**
   Dashed labels spanning their bars.
3. **Bars**
   Chord and its notes; the playing bar lights.
4. **TAB**
   Scrolls sideways; the sounding token lights.
5. **Notes**
   Four short cards explaining each section.

9 · JAM

Loop a backing track and improvise

Data: `JAM_TRACKS`, `jamExample()`, `JAM_IDEAS`.



1. **Tracks**
   Four buttons (`aria-pressed`). Switching while playing restarts on the new track.
2. **Current chord**
   Large display type, with "next: ..." beside it.
3. **Beat dots**
   Four dots, the current beat in `--color-next` (`beat_<n>` highlights).
4. **Start/Stop**
   Loops until stopped. Anything else that plays stops the jam.
5. **Tempo**
   60–140 BPM. Applies from the next loop.
6. **Try this**
   Five practice prompts.

"Show scale on fretboard" opens the popup on the track's scale and box, with its chords as pills. If a chord pill is on, its rings follow the chord that is playing.

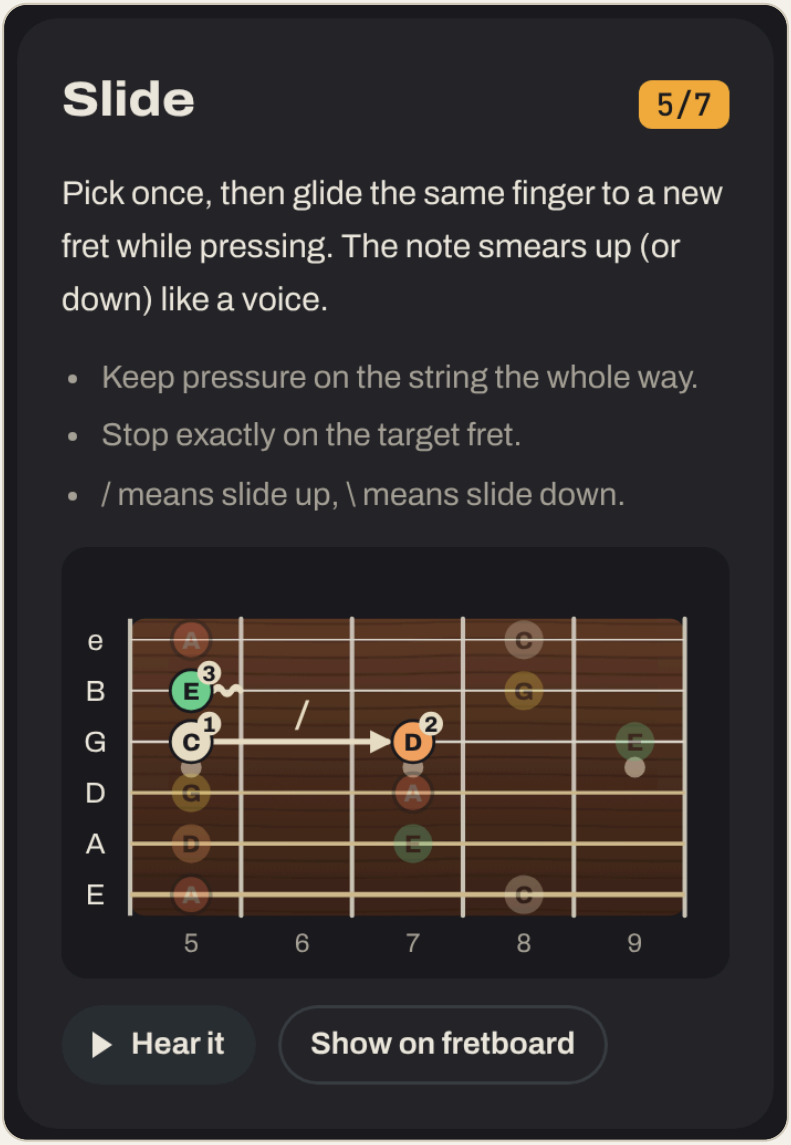
10 · PHONE (390 PX)

Popup and cards

The popup is centred near the top and stays draggable. The neck scrolls inside it. Cards stack to one column.



Fretboard map



Technique card

1 **Neck**
Scrolls sideways inside the popup; it opens scrolled to the box or lick. The "Drag" and "Chord tones" labels hide under 520px.

11 · ACCEPTANCE

Done when

The same checklist is in `cursor-prompt.md`.

1. /learn/solo renders without errors; no horizontal page scroll at 390 px; linked from the Learn page and the menu.
2. All 20 licks play through the app's lead voice (`playLead` + `articulation`) with their backing, and light their markers and TAB in sync, in the card and the popup.
3. Fretboard map: box row, chord-tone pills with rings and dashed outsiders, lick layer with badges and glyphs, Play/Stop and Clear lick. Existing uses of the popup (progress pages) are unchanged.
4. Box drawn in the lick's octave when a lick is shown.
5. Targeting A/B, full solo with sections, jam loop with tempo, current chord and beat dots.
6. Only one thing plays at a time; leaving the page stops audio.
7. Keyboard access with visible focus; Escape closes the popup; reduced motion removes scaling.
8. Tests: `soloGuide.test.ts` passes with the full suite; new tests for the neck props (box fade, rings, lick markers) and a Playwright spec (open popup, play a lick, switch box, run and stop the jam, phone layout).